

Doc. no.: P0509R1
Date: 2016-10-10
Project: Programming Language C++
Audience: Library Working Group
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Updating “Restrictions on exception handling”

GB 41 says of [res.on.exception.handling] that *“the “any other function” sentence in p4 contradicts the restriction placed in p2.”* Discussion in LWG revealed inconsistent interpretations of the text.

Proposed resolution: rewrite [res.on.exception.handling] to avoid ambiguity in which “other functions” are covered. Avoid saying “Throws clause”, as “clause” has another meaning in the Standard; say “Throws: paragraph” consistently. While here, address GB 42 by turning a “should” footnote into normative encouragement.

Drafting note: Please apply these edits after those for P0003.

1 Any of the functions defined in the C++ standard library can report a failure by throwing an exception of a type described in its **Throws:** paragraph, or of a type derived from a type named in the **Throws:** paragraph that would be caught by an exception handler for the base type. ~~An implementation may strengthen the exception specification for a non-virtual function by adding a non-throwing noexcept specification.~~

2 Functions from the C standard library shall not throw exceptions¹⁸⁶ except when such a function calls a program-supplied function that throws an exception.¹⁸⁷

3 Destructor operations defined in the C++ standard library shall not throw exceptions. Every destructor in the C++ standard library shall behave as if it had a non-throwing exception specification.

4 Functions defined in the C++ standard library that do not have a **Throws:** paragraph but do have a potentially-throwing exception specification may throw implementation-defined exceptions.¹⁸⁸ Implementations should report errors by throwing exceptions of or derived from the standard exception classes (18.6.3.1, 18.8, 19.2). ~~Any other functions defined in the C++ standard library that do not have an exception specification may throw implementation-defined exceptions unless otherwise specified.⁴⁸⁸ An implementation may strengthen this implicit exception specification by adding an explicit one.⁴⁸⁹~~

5. An implementation may strengthen the exception specification of a non-virtual function by adding a non-throwing exception specification.

GB 42:

“The word 'should' makes footnote 188 sound like normative encouragement, if not an actual mandate.”

Proposed resolution: make it normative encouragement, per the re-write of [res.on.exception.handling] for GB41 above. Remove it from the footnote.

188) In particular, they can report a failure to allocate storage by throwing an exception of type `bad_alloc`, or a class derived from `bad_alloc` (18.6.3.1). ~~Library implementations should report errors by throwing exceptions of or derived from the standard exception classes (18.6.3.1, 18.8, 19.2).~~

Acknowledgements

My thanks to Alisdair Meredith for assisting in understanding the effects of P0003 on this text.